01 / PROBLEM AND CONTRIBUTION

VeriChip verification

Digital Verification and Testing | Spring 2026 | Collaborative verification

RESULT AT A GLANCE

Built a self-checking SystemVerilog verification workflow for ALU operations and four controller states, supported by directed plans, assertions, cross-coverage, and VCS/Verdi debugging.

The engineering question

How can directed and constrained-random stimulus expose arithmetic and control corner cases?

Situation

The VeriChip ALU and controller have arithmetic, invalid-command, and error-state corner cases. A verification plan establishes the expected behavior before stimulus and checkers are applied.

Task

Translate the ALU and state-machine scenarios into self-checking directed and constrained-random tests, assertions, and coverage observations.

My contribution

SystemVerilog verification work; the ALU test plan was prepared with Akshith Alwala.

Tools and concepts

SystemVerilog / Assertions / Constrained-random tests / VCS / Verdi

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Organized ALU and state-machine scenarios around expected results, command handling, and state transitions.

02 / Implementation step

Used self-checking SystemVerilog tests, assertions, and cross-coverage, with VCS simulation and Verdi waveform debugging.

From plan to stimulus

The ALU and state-machine plans organize commands, expected values, invalid operations, and state transitions. These expectations guide stimulus and checking; a planning row alone is not evidence that a simulation passed.

Architecture / workflow

Conceptual summary of the documented workflow; not a simulation result.

01	Test stimulus
02	ALU / state machine
03	Assertions
04	Expected-value checks

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Controller states	4

Reported result

The illustrated brief shows selected ALU and state-machine checks and their expected values. It serves as a readable verification-plan example alongside the reviewed testbench.

Several views of correctness

Self-checking tests compare behavior, assertions express properties, and cross-coverage tracks combinations of interest. VCS/Verdi provide simulation and debug views, while achieved coverage still requires an actual report.

Scope and limitations

Expected values in a test plan are not measured pass counts or achieved coverage. No coverage percentage or bug count is inferred from the planning tables.

Possible next steps - proposed, not completed results

- Attach representative simulation logs and achieved coverage reports to substantiate completed regressions.
- Document the link from each planned corner case to its test or assertion.

Source material: ALU Test plan p. 6; SM_tstplan p. 3; reviewed SystemVerilog testbench; approved VeriChip brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.